How-To Library

Informatica

Performance tuning and best practices for Google Cloud Storage V2 Connector

© Copyright Informatica LLC 2025. Informatica, Informatica Cloud, and the Informatica logo are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.html.

Abstract

Supported Versions

- Informatica® Cloud Data Integration

Table of Contents

Abstract. 0

Overview. 2

Performance tuning areas. 2

Google Cloud Storage best practices. 3

Tune the Secure Agent. 3

JVM heap size. 3

Configure Secure Agent concurrency. 4

Tune the Google Cloud Storage mapping. 4

Compress files. 4

Tuning the number of parallel upload threads. 4

Overview

Performance tuning is an iterative process in which you analyze the performance, use guidelines to estimate and define parameters that impact the performance, and monitor and adjust the results as required.

This document describes the key hardware, database, Google Cloud Storage, Secure Agent, and Informatica mapping parameters that you can tune to optimize the performance of Google Cloud Storage V2 Connector. This document also includes case studies that involve the tuning specifications required when you configure a mapping to write data to Google Cloud Storage from a flat file using the Secure Agent. The graphs constructed from the case studies illustrate the impact that the tuning has on Google Cloud Storage V2 Connector performance.

Note: The performance testing results listed in this article are based on observations in an internal Informatica environment using data from real-world scenarios. The performance of Google Cloud Storage V2 Connector might vary based on individual environments and other parameters even when you use the same data.

Performance tuning areas

You can optimize the performance of a read or write operation when you use Google Cloud Storage V2 Connector by tuning the following parameters:

- Google Cloud Storage
- Secure Agent
- Google Cloud Storage mapping

Google Cloud Storage best practices

To optimize the performance, consider the following best practices for Google Cloud Storage:

- Ensure that the Secure Agent hosted on the Google Cloud Storage virtual machine is in the same region as the Google Cloud Storage dataset and the Google Cloud Storage bucket.
- Choose the correct Google Cloud Storage virtual machine instance type based on your requirements.

Tune the Secure Agent

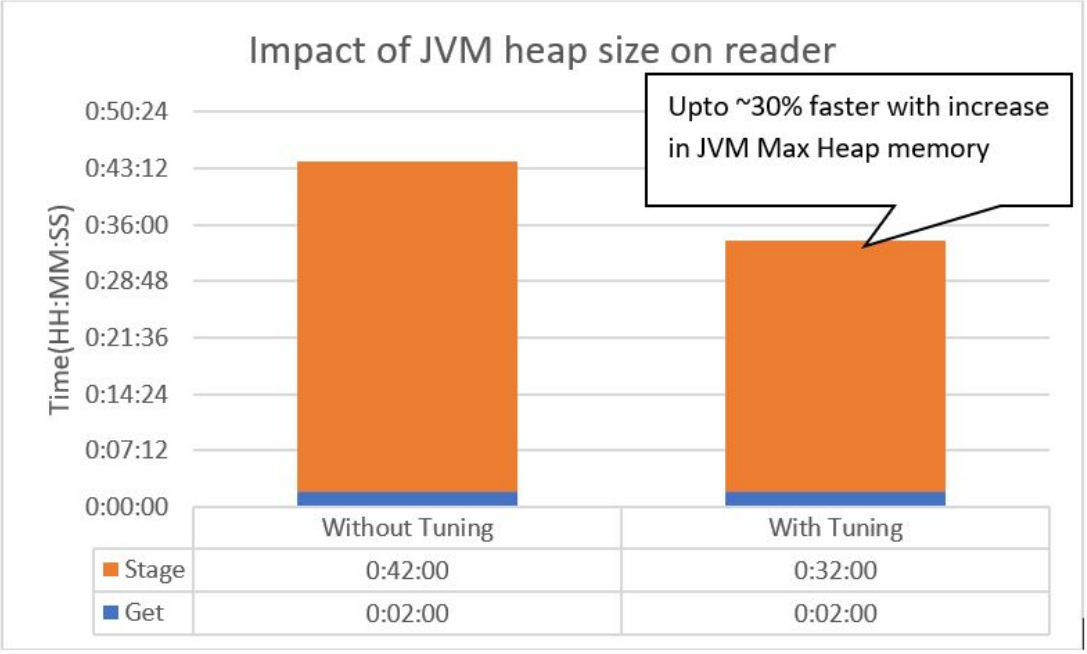
You can tune the JVM heap size and staging optimization process limit for the Secure Agent to optimize the performance of a Google Cloud Storage read or write operation.

JVM heap size

You can specify the amount of memory that the Secure Agent uses for each pmdtm process. Default is 64 MB.

If you use the default JVM heap size, the mapping might fail with an out-of-memory error. You can increase the JVM heap size to improve the performance of the read and write operations.

The following graph shows the results and the recommended values for setting the JVM maximum heap size:



For Google Cloud Storage V2 mappings and mapping tasks that read data from Google Cloud Storage, specify a JVM memory of -Xmx256m.

If you already configured a heap size value that is higher than 256 MB in the JVM options, do not change it to -Xmx256m.

Perform the following steps to configure the JVM memory:

1. In Administrator, select the Secure Agent listed on the **Runtime Environments** tab.
2. Click **Edit**.
3. In the **System Configuration Details** section, select **Data Integration Service** as the service and **DTM** as the type.

4. Edit the **JVMOption** property, and enter *-Xmx256m*.
5. Click **Save**.

Configure Secure Agent concurrency

By default, the Secure Agent processes two concurrent mapping tasks.

If there are more than two concurrent tasks, additional tasks are queued and then scheduled for execution when a slot becomes available. This can cause the capacity of the Secure Agent machine to be underutilized.

To achieve better utilization of the CPU capacity of the Secure Agent machine and achieve a higher degree of concurrency, you can set the `maxDTMProcesses` custom property for the Data Integration Server to the number of parallel tasks.

To configure the Secure Agent concurrency, perform the following steps:

1. In **Administrator**, open the **Runtime Environments** page.
2. Select the Secure Agent to view its details, and then click **Edit** on the **Secure Agent** page.
3. Specify the following details:
   - Service. Choose **Data_Integration_Server**.
   - Type. Choose **Tomcat**.
   - Name. Enter `maxDTMProcesses`.
   - Value. Enter the Secure Agent concurrency value.
4. Click **Save**.

For example, if you set the `maxDTMProcesses` property to 16, the Secure Agent allows 16 mapping tasks to run simultaneously.

Note: Each mapping task spawns its own JVM and reserves the specified JVM heap from the physical memory.

Tune the Google Cloud Storage mapping

You can configure the following properties to optimize the Google Cloud Storage mapping performance:

- File compression
- Tuning the number of parallel upload threads

Compress files

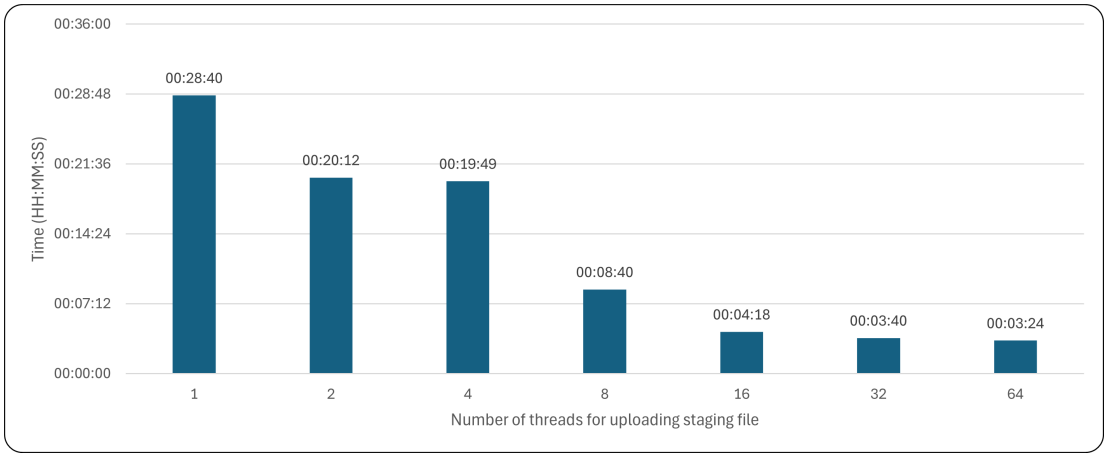
If the Secure Agent and Google Cloud Storage are configured in different regions and there is a network latency, you can enable compression to compress the staged file to reduce the transfer time when you read data from or write data to Google Cloud Storage.

Tuning the number of parallel upload threads

You can optimize the performance of uploading a staging file when you write data to a Google Cloud Storage target.

Informatica recommends selecting the **Use Parallel Composite Upload** target advanced property to optimize the performance of uploading the staging file.

The following graph shows the impact of upload performance relative to the number of parallel threads used for Google Cloud Storage upload:



Authors

Sudheesh Kumar E

Mridul Mathew